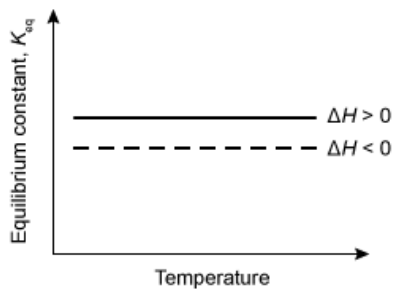


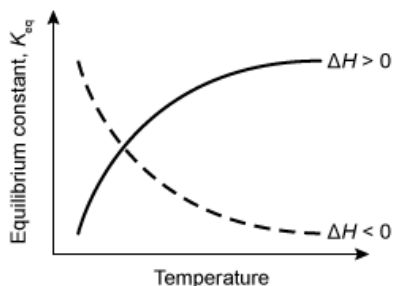
Which graph best represents the relationship between temperature and the equilibrium constant K_{eq} of an exothermic and endothermic reaction?

☐ A.



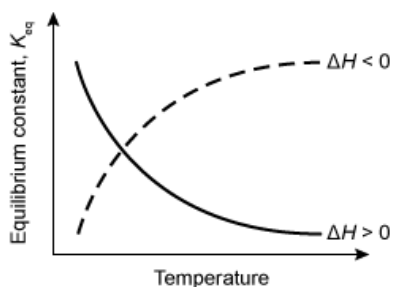
[5%]

☒ B.



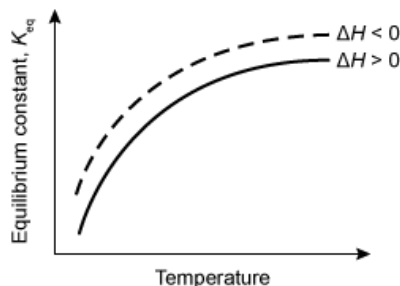
[58%]

☐ C.



[31%]

☐ D.

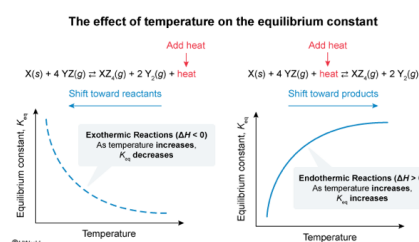


[4%]

Correct

58% answered correctly

Explanation:



A reaction's **equilibrium constant**, K_{eq} , measures to what degree reactants are converted to products and is equal to the **equilibrium molar ratios** (or **partial pressures**) of products to reactants:

$$K_{eq} = \frac{[\text{Products}]^{\text{coefficients}}}{[\text{Reactants}]^{\text{coefficients}}}$$

The **temperature dependence of K_{eq}** depends on whether a reaction is **exothermic ($\Delta H^\circ < 0$)** or **endothermic ($\Delta H^\circ > 0$)**. An exothermic reaction releases heat (ie, heat is a product), and increasing the temperature (ie, adding heat) causes the reaction to shift toward reactants (ie, K_c decreases) to compensate for the heat gain, in accordance with **Le Châtelier's principle**. Alternatively, an endothermic reaction absorbs heat (ie, heat is a reactant), and the reaction will shift toward products (ie, K_c increases) with increasing temperature.

Therefore, a graph of K_{eq} versus temperature will show K_{eq} **decreasing** with **increasing** temperature for an exothermic reaction and K_{eq} **increasing** with **increasing** temperature for an endothermic reaction, as depicted in Choice B.

(Choice A) This graph suggests K_{eq} is temperature *independent*, which is *not* true.

(Choices C and D) As temperature increases, K_{eq} will *decrease* for the *exothermic* reaction and *increase* for the *endothermic* reaction.

Educational objective:

The equilibrium constant K_{eq} expresses the relationship between products and reactants at equilibrium. When temperature increases, K_{eq} decreases for exothermic reactions and increases for endothermic reactions.

Time spent: 0 sec
Subject:
General Chemistry

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System:
Thermodynamics, Kinetics & Gas Laws